Robust Quality Classification of Germinated Oil Palm Seeds under Real-World Distribution Shift

COMP3029 Computer Vision
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Abstract. Deep neural networks deployed in agricultural quality inspection can learn shortcuts invisible under controlled testing yet persist with full confidence under real-world distribution shift. This paper investigates this phenomenon in a ResNet-18 classifier for germinated oil palm seed grading, which achieves 0.9576 in-domain Macro-F1 yet degrades to 0.6700 under pose and scale shift. Grad-CAM forensic analysis identifies two failure patterns sharing a single root cause: exclusive reliance on pixel intensity contrast with no morphological reasoning. A targeted Morphological and Photometric Decoupling strategy is formulated to destroy this shortcut at training time, achieving F1 improvements of +28.26% on Batch-3 and +13.69% on Batch-2 while retaining in-domain performance. These results demonstrate that forensic shortcut identification is a more principled path to distributional robustness than architectural scaling.

Keywords: Shortcut learning · Distribution shift · Grad-CAM · ResNet-18 · Oil palm seed grading · Photometric decoupling · Covariate shift

1 Introduction

Automated quality grading of germinated oil palm seeds is an economically significant problem: germination viability, determined by the structural integrity of the plumule and radicle, dictates crop yield, and manual inspection at industrial scale is labour-intensive and subject to inter-rater variability. Deep convolutional neural networks offer a natural solution, yet field deployment exposes a tension between laboratory performance and real-world robustness.

Distribution shift can degrade models in two qualitatively different ways. In the first, covariate shift disrupts learned representations and produces detectable low-confidence failures. In the second, a model has learned a spurious correlation that holds across domain boundaries; it fails silently with full confidence. This paper identifies and addresses the second mode. Prior investigations on this dataset applied augmentation as general regularisation and pursued architectural complexity [3], yielding incremental gains without identifying the causal failure mechanism. Every design decision here is motivated by a specific forensic Grad-CAM observation, and the central research question is whether forensic identification of a specific shortcut, followed by its targeted elimination, can produce more robust generalisation than the strategies previously applied.

The baseline ResNet-18 follows standard transfer learning practice; the original contribution is the forensic diagnosis (Section 4) and the targeted shortcut elimination derived from it (Sections 5–7).

2 Baseline Model

2.1 Architectural Selection

A pre-trained ResNet-18 [1] with a binary fully connected classification head is selected as the sole baseline. Prior work reports ResNet-18 outperforms ResNet-50 under pose shift (80.0% vs. 71.8% accuracy on Batch-3), consistent with shortcut learning theory: deeper architectures encode more source-domain statistics and are more vulnerable to covariate shift [3, 4]. ResNet-18's simplicity ensures interpretability in Grad-CAM analysis. Retaining the single fully connected head rather than adopting a two-layer MLP [3] acts as a scientific control, ensuring metric recovery on Batch-2 and Batch-3 is attributable to robust feature representation rather than classifier capacity.

2.2 Training Protocol and Dynamics

The model is trained exclusively on Batch-1 using Adam (lr = 0.001), cross-entropy loss, batch size 64, over 10 epochs, with fixed random seed (42) and deterministic CUDA operations. Images are resized to 224×224; centre-cropping was avoided as it risks truncating germination tissue. Training loss converges to near-zero by Epoch 5 while validation loss diverges, confirming source-domain memorisation rather than morphological generalisation (Fig. 1).

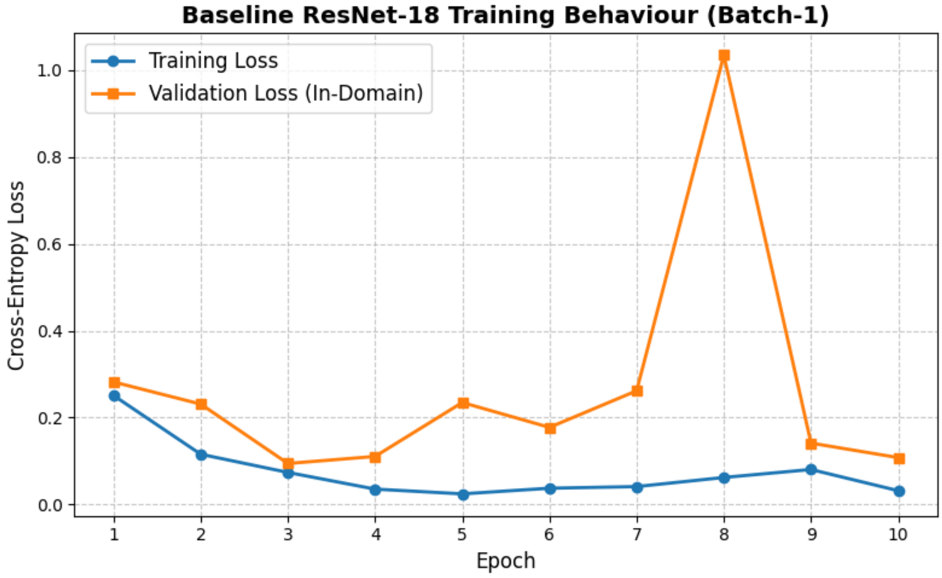

Fig. 1. Baseline training dynamics. Rapid training-loss convergence with diverging validation loss confirms source-domain memorisation.

3 Baseline Evaluation

The baseline is evaluated on all three batches using Accuracy, Macro-Precision, Macro-Recall, Macro-F1, and AUC-ROC. Macro-F1 is the primary metric for class-balanced assessment. For class set C = {GoodSeed, BadSeed}:

$$F_{1,macro} = \frac{1}{2}\sum_{c=0}^{1}\frac{2 \cdot P_c \cdot R_c}{P_c + R_c}$$

where P_c and R_c are per-class precision and recall. Batch-2 and Batch-3 images are cropped via ground-truth bounding boxes.

Table 1. Dataset composition. Individual seed counts are extracted from pre-segmented crops (Batch-1) and ground-truth bounding box annotations (Batch-2, Batch-3).

Batch	Role	Good-Seed	Bad-Seed	Total	Camera	Focal	Lighting	Distribution Shift
Batch-1 (Train)	Train	901	851	1,752	Samsung NX2000	40 cm	Lightbox	None (in-domain)
Batch-1 (Val)	Validation	201	200	401	Samsung NX2000	40 cm	Lightbox	None (in-domain)
Batch-2	Test	450	450	900	Canon IXUS 285 HS	26 cm	Fluorescent	Lighting shift
Batch-3	Test	605	593	1,198	Canon IXUS 285 HS	26 cm	Lightbox	Pose/scale shift

Table 2. Cross-domain baseline evaluation results.

Dataset	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Batch-1 (In-Domain)	95.76%	0.9577	0.9576	0.9576	0.9947
Batch-2 (Lighting Shift)	79.44%	0.8377	0.7944	0.7876	0.9039
Batch-3 (Pose/Scale Shift)	70.03%	0.8061	0.6974	0.6700	0.8806

The degradation is structured rather than random. The photometric shift in Batch-2 produces a 17.9% relative F1 decline; the combined geometric and hardware shift in Batch-3 produces a 30.2% decline. This two-fold difference implies geometric covariate shift is the dominant failure mode, an a priori prediction tested in Section 7. The confusion matrices (Fig. 2) reveal a systematic false-positive bias, consistent with a shortcut favouring the GoodSeed class.

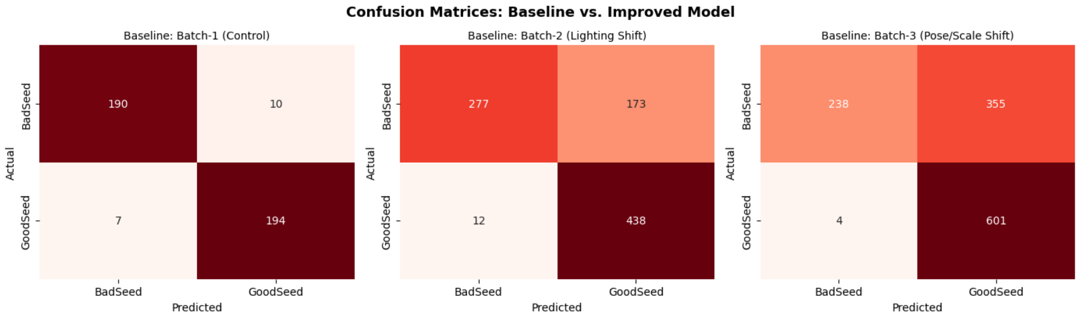

Fig. 2. Confusion matrices for the baseline model across all three batches. Systematic false-positive bias in Batch-2 and Batch-3 is consistent with a learned shortcut.

4 Systematic Error Analysis

Grad-CAM [2] visualisations are generated for the three highest-confidence misclassifications per shifted batch. Selecting on confidence targets the features the model finds most discriminative in the absence of the correct signal, exposing the learned shortcut most clearly.

4.1 Pattern 1: Presence Bias

Five false-positive cases across Batch-2 (Rows 1–2) and Batch-3 (Rows 1–3) show the model correctly localising germination tissue but remaining entirely insensitive to its biological quality. Defective sprouts (multi-branched, stunted, or irregularly structured) are predicted as GoodSeed at 100.0% confidence in all five cases, with Grad-CAM heatmaps saturating uniformly on the white tissue mass. The model's decision reduces to a single binary question: does a high-contrast white or yellow pixel cluster exist in the image? If yes, the prediction is GoodSeed, regardless of structural arrangement or biological integrity. This is a representational failure, not a localisation failure: the model attends to the correct anatomical region but has never learned what healthy germination actually looks like.

4.2 Pattern 2: Shortcut Collapse

The complementary failure mode is observed in Batch-2 Row 3, where a genuine GoodSeed is predicted as BadSeed at 100.0% confidence. The seed coat here exhibits a strongly striped texture whose tonal values closely approach those of the white sprout, suppressing the contrast differential the shortcut depends on. The Grad-CAM heatmap consequently fires across the seed coat stripes rather than localising on the sprout, and with no morphological fallback, the model defaults to BadSeed.

Pattern 2 is the logical complement of Pattern 1: the shortcut fires on any high-contrast white tissue regardless of quality (Pattern 1, five cases) and collapses when seed coat texture disrupts the contrast it depends on (Pattern 2, one case). Across all five Pattern 1 cases, Grad-CAM heatmaps remain anchored to the germination region despite lighting change, hardware change, and pose variation, confirming the shortcut is domain-invariant. Standard domain adaptation is therefore unlikely to be effective, as the features are already aligned across domains but attend to the wrong signal.

Taken together, both patterns confirm a single root cause: exclusive reliance on pixel intensity contrast with no morphological fallback, and a training intervention that eliminates this shortcut is the only viable remediation.

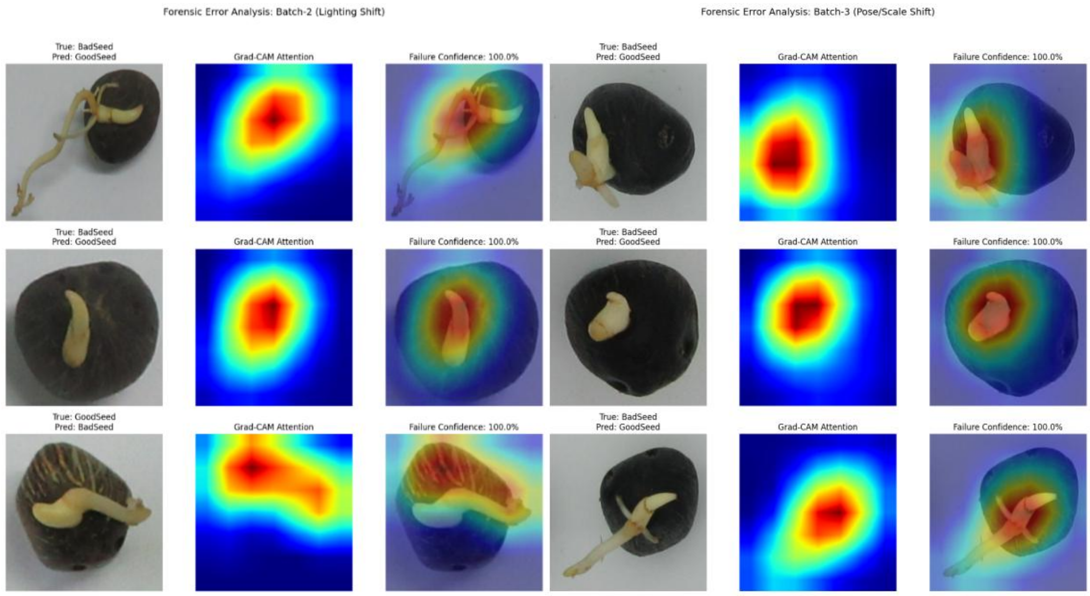

Fig. 3. Grad-CAM analysis for Batch-2 (left) and Batch-3 (right). Rows 1 and 2 of Batch-2 and all rows of Batch-3: presence bias. Batch-2 Row 3: shortcut collapse. All cases share the same representational root cause.

5 Hypothesis

The forensic evidence from Section 4 leads to the following explicit, falsifiable hypothesis.

Hypothesis:

Performance degradation under distribution shift is causally attributable to the model's exclusive reliance on pixel intensity contrast between germination tissue and seed coat as its sole discriminative signal. Because this shortcut is biologically invalid and domain-invariant by construction, it produces confident misclassifications under both lighting and pose shift: misfiring on defective tissue when contrast is present and collapsing when seed coat texture suppresses the contrast differential. Targeted elimination during training is expected to recover cross-domain performance while retaining in-domain accuracy.

The shortcut must be actively destroyed by constructing an augmented distribution in which pixel intensity is no longer a statistically reliable class signal. A pipeline incorporating geometric invariance augmentation (multi-scale cropping, rotation) and photometric decoupling (colour jitter, random grayscale) is expected to compel the model to develop structural and textural representations invariant to both domain shift and contrast variation.

Three falsifiable a priori predictions follow. (i) Macro-F1 must improve by at least +10% on both Batch-2 and Batch-3. (ii) The improvement on Batch-3 must exceed that on Batch-2, since geometric shift is the dominant failure mode; a general regularisation effect would not produce this asymmetric ordering, and a result where Batch-2 improves more would directly falsify the hypothesis. (iii) Batch-1 performance must be retained, confirming shortcut elimination does not destroy in-domain discriminative capacity. Retaining ResNet-18 as a scientific control ensures metric recovery is attributable to feature representation rather than model capacity.

6 Hypothesis-Driven Improvement

The single targeted modification is a Morphological and Photometric Decoupling augmentation pipeline in which each transform eliminates one forensically identified failure mechanism. This retains ResNet-18 as a scientific control, ensuring performance recovery is attributable to robust feature representation.

6.1 Augmentation Design

Let x denote an original training image. The augmented sample is:

$$\tilde{x} = (T_{photo} \circ T_{geo})(x)$$

where T_{geo} applies geometric invariance transforms (scale, pose) and T_{photo} applies photometric decoupling transforms, each targeting a specific failure from Section 4.

Table 3. Augmentation pipeline. Each transform maps forensically to one identified failure mechanism.

Intervention	Target Failure	Implementation
Scale Invariance	Focal distance shift (40 cm to 26 cm)	RandomResizedCrop(scale=0.6–1.0)
Pose Invariance	3D seed orientation (Batch-3)	RandomRotation(180°) + RandomFlips
Photometric Decoupling	Intensity contrast shortcut (both batches)	RandomGrayscale(p=0.2) + ColorJitter(b=0.4, c=0.4, s=0.2)
Sensor Simulation	Canon camera noise profile	GaussianBlur(kernel=3, σ=0.1–2.0)
Optimisation	Augmentation-induced gradient noise	Adam (lr=1e−4), CosineAnnealingLR [6], wd=1e−5

The photometric decoupling parameters are deliberately aggressive. RandomGrayscale at $p = 0.2$ intermittently removes all colour information, forcing discriminative capacity in a regime where the intensity shortcut is entirely absent. ColorJitter at 0.4 randomises the luminance relationship constituting the shortcut. A weaker corruption would train the model to hedge rather than fully abandon the intensity cue.

6.2 Validation-Aware Checkpointing

Heavy augmentation [5] can induce augmentation memorisation, where the model overfits to the statistical regularities of the augmented distribution rather than to the original data. The model is therefore evaluated against the unaugmented Batch-1 validation set at every epoch, retaining only the minimum validation loss weights. This minimum occurs at Epoch 14 (Fig. 4); all Section 7 results reflect these weights.

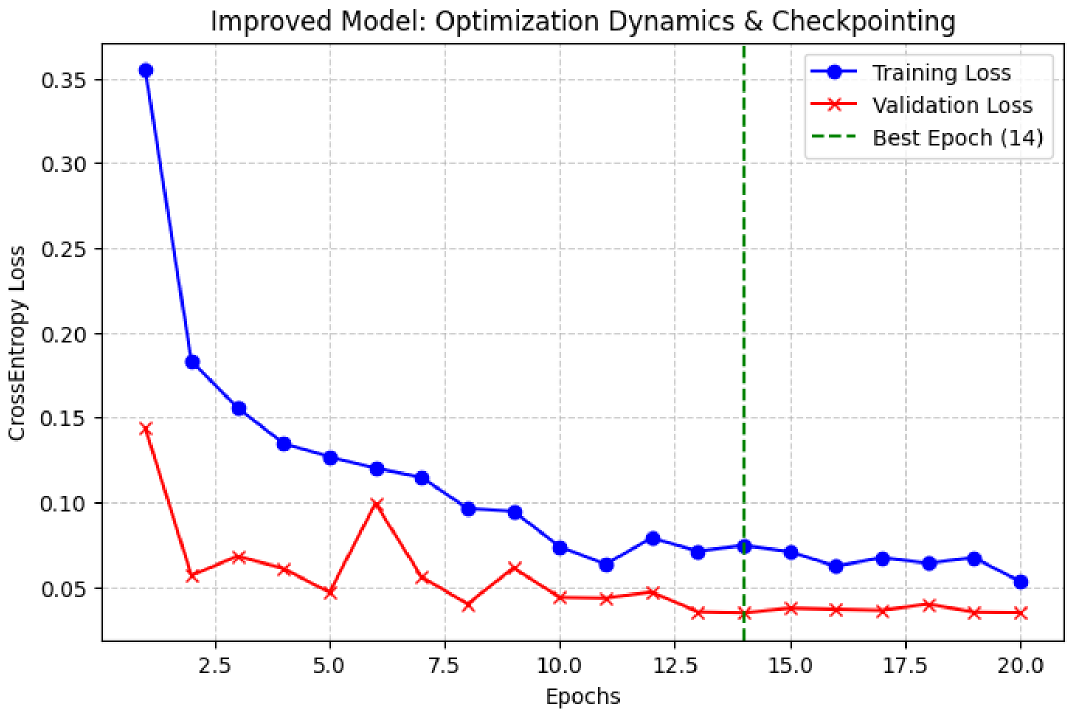

Fig. 4. Improved model training dynamics over 20 epochs. Validation loss is volatile, characteristic of augmentation-heavy training. Epoch 14 is identified as the point of minimum validation loss.

7 Experimental Results

The improved model is evaluated using the identical protocol, data loaders, and bounding-box cropping as Section 3. The sole independent variable is the training pipeline; architecture, random seed, and evaluation methodology are held constant.

Table 4. Comparative F1 and AUC-ROC results. The improvement gradient (Batch-3 > Batch-2 > Batch-1) is structurally consistent with the a priori hypothesis.

Dataset	Baseline F1	Improved F1	Absolute Δ	Relative Δ	Baseline AUC	Improved AUC
Batch-1 (In-Domain)	0.9576	0.9900	+0.0324	+3.39%	0.9947	0.9997
Batch-2 (Lighting Shift)	0.7876	0.8955	+0.1079	+13.69%	0.9039	0.9675
Batch-3 (Pose/Scale Shift)	0.6700	0.8594	+0.1894	+28.26%	0.8806	0.9366

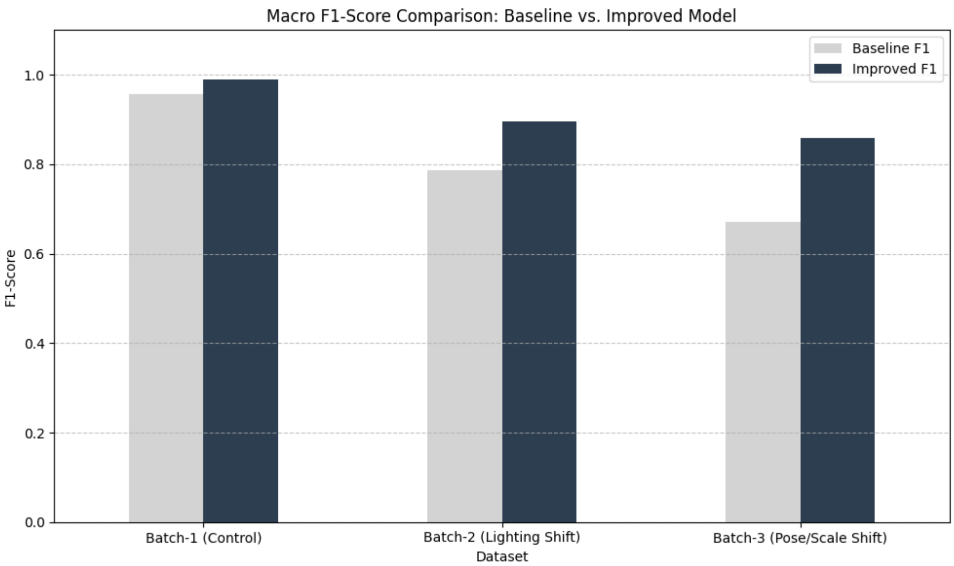

Fig. 5. Macro-F1 comparison by batch. The asymmetric improvement gradient confirms the intervention worked for the predicted reason.

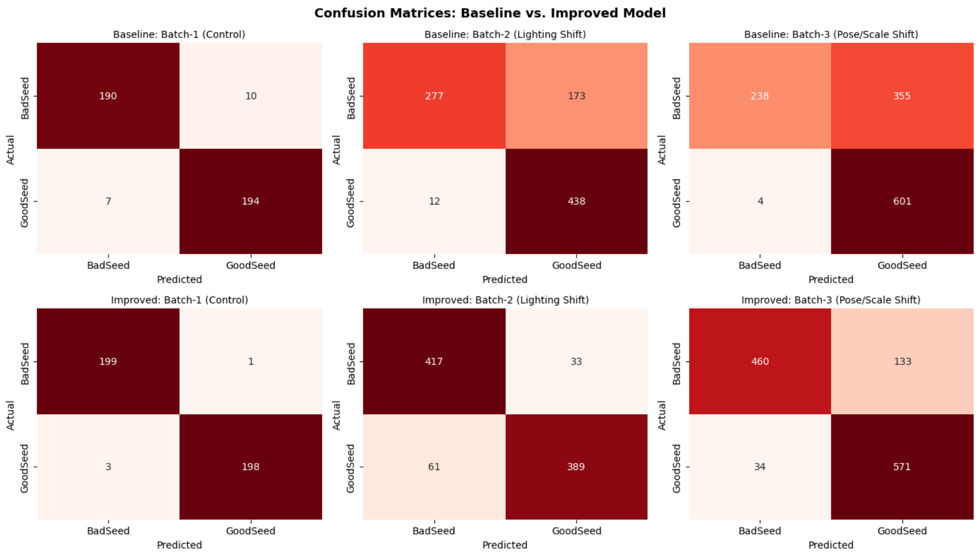

Fig. 6. Confusion matrices: baseline (top) and improved model (bottom). The systematic false-positive bias is substantially corrected.

All three falsifiable predictions are confirmed. The minimum +10% F1 threshold is exceeded: +13.69% on Batch-2 and +28.26% on Batch-3. The Batch-3 improvement is approximately twice that of Batch-2, precisely matching the a priori structural prediction. A general regularisation effect would not produce this asymmetric ordering; the result therefore rules out the alternative explanation.

Batch-1 F1 improved marginally (+3.39%), confirming no catastrophic forgetting. The AUC improvements (Batch-3: 0.8806 to 0.9366; Batch-2: 0.9039 to 0.9675) are threshold-independent and indicate genuine restructuring of the model's probability landscape. The marginal Batch-1 gain is analytically consistent: by eliminating the intensity shortcut, the model is compelled to develop morphological features that happen to be marginally more discriminative even in-domain.

8 Discussion

8.1 Implications

The central finding is that the shortcut survives rather than collapses under domain shift, challenging the assumption that covariate shift degrades performance by disrupting learned representations. When a shortcut is calibrated to a physical property of the object, specifically spectral contrast between tissue and seed coat, it can be domain-invariant by construction. The two-directional failure pattern, comprising five presence-bias false positives and one shortcut-collapse false negative, confirms the shortcut is the sole decision mechanism with no morphological fallback. Shortcuts anchored to object-intrinsic properties rather than environmental conditions [4] are particularly dangerous because standard distribution shift monitoring and domain adaptation are unlikely to expose or address them.

8.2 Limitations

Three limitations constrain generalisability. First, shortcut elimination is partial: a residual gap of 0.1406 F1 from perfect Batch-3 classification remains. Grad-CAM reveals spatial attention but cannot fully characterise learned feature semantics; the intensity contrast interpretation is strongly indicated but not definitive. Second, the single-viewpoint protocol is inherently ill-posed for assessing radicle development, and no augmentation strategy can fully compensate for missing three-dimensional geometric information. Third, the training set of 1,752 images limits coverage of deployment variation, and augmentation parameters were not validated against a held-out pose distribution.

8.3 Future Directions

Consistency regularisation, implemented as a loss term penalising prediction disagreement between differently augmented views of the same seed, would extend shortcut elimination into the loss function, providing an explicit gradient signal against shortcut-based predictions. A lightweight spatial attention module such as CBAM [7] targeting the germination zone would enable finer intra-sprout discrimination, directly addressing Pattern 1. Multi-view imaging or depth sensing would resolve the single-viewpoint constraint by providing the three-dimensional geometric information that two-dimensional projection discards.

9 Conclusion

This investigation demonstrates that high in-domain performance does not guarantee distributional robustness. The ResNet-18 baseline, despite achieving 0.9576 Macro-F1 on Batch-1, collapses to 0.6700 under pose shift through a specific mechanism: exclusive reliance on pixel intensity contrast with no morphological reasoning. This shortcut is domain-invariant, producing silent 100.0% confidence misclassifications across both shifted batches without triggering any performance warning in the source domain.

A targeted Morphological and Photometric Decoupling strategy achieves +28.26% F1 on Batch-3 and +13.69% on Batch-2. The asymmetric improvement gradient, larger on the more severely shifted domain, provides structural confirmation that the intervention worked for the predicted reason. Retaining ResNet-18 as a scientific control isolates data representation as the causal variable, confirming that understanding why a model fails is a more effective investment than scaling what fails.

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